

MIGC-CMamba: Cross-Domain Mamba with Multi-Scale Imaging and Granular-Ball Computing for Traffic Flow Prediction

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Abstract

With the increasing relevance of web mining and content analysis in uncovering mobility patterns from large-scale online data, traffic flow prediction plays a crucial role in proactive urban planning and enhancing the responsiveness of intelligent transportation systems. However, existing traffic flow prediction methods often fail to explicitly capture correlations in continuous multivariate sequences that are naturally suited for trend and periodic pattern extraction by vision models and rely on fixed spatial graphs that neglect the cognitive advantages of granular-ball structures, which limits their ability to model interactions and strengthen spatiotemporal dependencies. To address these challenges, this paper proposes a Cross-domain Mamba framework that integrates Multi-scale Imaging and Granular-ball Computing for traffic flow prediction (MIGC-CMamba). First, a multi-scale sequence imaging method is presented, which converts the original time series into image modality and leverages MambaVision to capture both local and global dependencies. Second, a multi-granularity spatial graph is constructed via granular-ball clustering, which balances global trend representation and local detail preservation. Third, a cross-domain enhancement mechanism adaptively integrates temporal and spatial domains, strengthening spatiotemporal dependencies. Lastly, extensive experiments demonstrate superior performance over state-of-the-art baselines, highlighting how vision-based imaging, cognition-inspired granular-ball modeling, and content-aware mining jointly advance the modeling of spatiotemporal dependencies in traffic flow prediction.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; • **Information systems** → **Spatial-temporal systems**.

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WWW '26, Dubai, United Arab Emirates.
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ACM ISBN 979-8-4007-2307-0/2026/04
<https://doi.org/10.1145/3774904.3792359>

Keywords

Traffic Flow Prediction; Multi-Scale Imaging; Granular-Ball Computing; Mamba; Cross-Domain Enhancement

ACM Reference Format:

Wenxia Chang, Chao Zhang, Wentao Li, and Deyu Li. 2026. MIGC-CMamba: Cross-Domain Mamba with Multi-Scale Imaging and Granular-Ball Computing for Traffic Flow Prediction. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3774904.3792359>

1 Introduction

In the era of rapidly evolving information technology, vast amounts of unstructured traffic-related data have emerged in cyberspace, including social media posts, news reports, navigation records, and user reviews. Web mining and content analysis techniques make it possible to transform these heterogeneous sources into structured knowledge, revealing travel behaviors, road usage patterns, and emerging traffic events [14]. By integrating such insights with traditional sensing data, traffic flow prediction can be more effectively advanced, providing crucial support for alleviating congestion, optimizing resource allocations, and improving travel efficiencies in intelligent transportation systems [21, 36].

A key challenge in traffic flow prediction involves accurately capturing complex spatiotemporal dynamics. Traditional sequence models such as recurrent neural networks (RNNs) and long short-term memory (LSTM) capture short-term temporal dependencies but experience vanishing gradients and struggle to model long-range modeling [23]. To overcome these drawbacks, Transformers have become mainstream by leveraging self-attention, which enables parallel computation and effectively captures long-range temporal dependencies, thereby improving training efficiency and predictive accuracy [29]. However, Transformers face two critical issues. First, their quadratic computational complexity $O(N^2)$, where N represents the input sequence length, results from the pairwise attention operation between all tokens. This leads to high computational and memory costs, severely limiting scalability in long-sequence modeling. Second, their ability to effectively capture localized spatial or temporal dependencies remains limited [15]. In contrast, state space models (SSMs) reduce complexity to $O(N)$ via linear recurrence, making them highly efficient for long sequences, yet their linear nature constrains their ability to model nonlinear interactions and complex spatial correlations that are essential in

traffic data [22]. To address this issue, the recently proposed Mamba architecture introduces a dynamic selection mechanism combined with attention mechanism, enabling more expressive spatiotemporal fusion while preserving linear complexity [10]. Nevertheless, even with these improvements, sequence models including Mamba struggle to explicitly capture multivariate correlations and long-term periodicity. Since traffic flow exhibits strong trends and peaks, image-based representations provide a promising complementary solution, as they naturally encode continuous temporal patterns and allow vision models to leverage pre-trained knowledge of structural regularities. Building on this intuition, MambaVision extends Mamba into the visual domain, where multi-scale imaging enhances spatiotemporal representation learning, making it particularly suitable for traffic flow prediction, which requires the accurate modeling of both spatial dependencies and long-range temporal dynamics [9].

Although recent advances have enhanced spatiotemporal representation learning, traffic flow prediction still faces several challenges.

- **Single-scale temporal modeling fails to capture multi-scale patterns.** Modeling multi-scale temporal patterns is difficult, as traffic exhibits both rapid fluctuations and long-term periodic trends.
- **Fixed-topology spatial modeling struggles to represent multi-granularity dependencies.** Spatial dependencies span multiple granularities, from local segments to regional clusters, demanding adaptive clustering beyond fixed topologies.
- **Independent spatiotemporal modeling lacks effective cross-domain interaction.** Effective prediction requires not only separate temporal and spatial modeling but also their dynamic interaction to capture cross-domain complementarities.

To this end, this paper proposes three modules: multi-scale time series imaging for temporal dynamics, granular-ball clustering for adaptive multi-granularity spatial modeling, and a cross-domain complementary enhancement module for fusing spatial and temporal features.

Firstly, traffic flow exhibits complex temporal patterns characterized by rapid short-term fluctuations and recurring long-term periodicity, posing significant challenges for effective temporal modeling, as illustrated in Fig. 1 (a,b) [18, 25]. To address this limitation, multi-scale modeling captures patterns across multiple time scales, enabling the representation of both fine-grained local variations and overarching long-term trends. Furthermore, images naturally provide a structured representation of traffic data, which facilitates the extraction of local patterns as well as global trends. Taking advantage of this structured representation, MambaVision employs structured SSMS to efficiently capture long-range temporal dependencies, making it well suited for multi-scale temporal feature extraction. Accordingly, this paper designs multi-scale time series imaging to represent temporal dynamics and applies MambaVision to extract multi-modal temporal features.

Secondly, traffic networks inherently exhibit hierarchical spatial structures, where local roads interact within subregions and regional flows reflect broader connectivity patterns, as illustrated

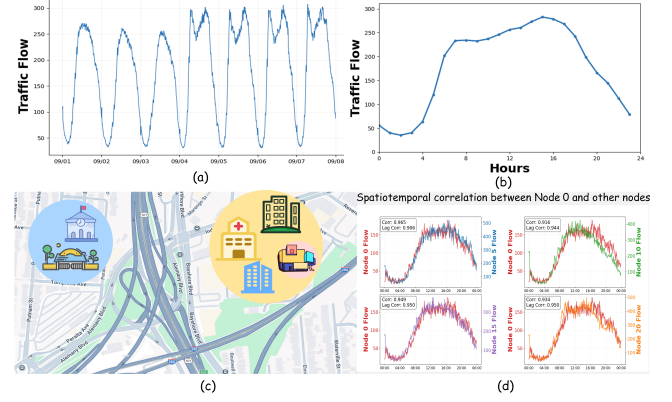


Figure 1: Traffic flow prediction modules. (a) visualizes weekly traffic periodicity, (b) shows daily flow fluctuations, (c) presents multi-granularity spatial topology, and (d) illustrates spatiotemporal dependencies.

in Fig. 1 (c) [34]. Effectively modeling such multi-granularity dependencies requires adaptive representations that can balance local detail with global coherence. The granular-ball framework naturally provides this advantage as it adaptively groups data points into hierarchically organized balls based on similarity, allowing flexible granularity control without relying on predefined thresholds or fixed topology. Accordingly, this paper constructs the granular-ball clustering method to adaptively aggregate traffic network nodes across multiple granularities, preserving spatial topology while reducing graph complexity.

Thirdly, temporal and spatial dependencies in traffic flow exhibit inherently complementary characteristics, where temporal features reflect sequential evolution and periodic patterns, while spatial features capture structural correlations and regional interactions [12]. As illustrated in Fig. 1 (d), the spatial correlation of traffic flow across different nodes further demonstrates the coupled nature of spatiotemporal dependencies. Effectively integrating these two domains is crucial for modeling complex spatiotemporal dynamics. The cross-domain complementary enhancement mechanism naturally facilitates effective information integration by enabling bidirectional interaction and mutual reinforcement among heterogeneous feature domains. Accordingly, this paper suggests a cross-domain complementary enhancement module that dynamically fuses temporal and spatial feature channels, thereby strengthening inter-domain interactions and improving the overall modeling of spatiotemporal dependencies.

In summary, this paper proposes the **MIGC-CMamba** model, which integrates multi-modal with multi-granularity information for spatiotemporal modeling, effectively enhancing feature interactions and improving the representation of complex spatiotemporal relationships. The key innovations of this paper are summarized as follows:

- **Multi-scale sequence imaging enhances multi-range temporal modeling.** A multi-scale sequence imaging method is designed, leveraging the advantages of image processing

in extracting both local and global features for traffic flow prediction.

- **Granular-ball clustering enables adaptive spatial modeling.** A multi-granularity spatial graph construction method based on granular-ball clustering is constructed. To the best of our knowledge, granular-ball computation is employed in traffic flow prediction for the first time.
- **Cross-domain fusion strengthens spatiotemporal interaction.** A cross-domain feature complementary enhancement method is suggested, enabling dynamic interaction and collaborative enhancement of spatiotemporal features.
- **Extensive experiments demonstrate strong baseline performance.** Experiments are conducted on four real-world datasets, including highway traffic data from the California Performance Measurement System (PeMS), namely PeMS03, PeMS-Bay, and METR-LA, and an urban dataset (JiNan). The proposed model consistently outperforms baselines, achieving a 33.94% reduction in RMSE on PeMS-Bay.

2 Problem Definition

This section focuses on two key concepts in traffic prediction, i.e., the graph representation of the traffic network and historical-data-based traffic flow prediction.

Definition 1 (Traffic Network): The traffic network can be represented as a graph $G = (V, E, A)$, where $V = \{v_1, v_2, \dots, v_n\}$ is the set of nodes ($|V| = N$), each indicating a key location in the traffic network, such as intersections, transit stations, or sensors. $E \subseteq V \times V$ is the set of edges, representing the connections between nodes. $A \in \mathbb{R}^{N \times N}$ denotes the adjacency matrix, which represents the connection strengths among nodes.

With the traffic network defined above, the aim of traffic flow prediction can be formally stated.

Definition 2 (Traffic Flow Prediction): Traffic flow prediction focuses on predict the traffic flow for a future time period by analyzing historical traffic data. Given the traffic flow data $X \in \mathbb{R}^{T \times N \times C}$ for the past T time steps, where T represents the historical window length, N indicates the total nodes in the traffic network, and C denotes the number of node attributes. By learning a mapping function f , the model predicts traffic flow for the upcoming T' time steps.

$$[X_{t-T+1}, X_{t-T+2}, \dots, X_t] \xrightarrow{f} [Y_{t+1}, \dots, Y_{t+T'}]. \quad (1)$$

3 Methodology

Figure 2 illustrates the MIGC-CMamba model architecture. **Module (a)** takes raw traffic data as input and performs the initial preprocessing for subsequent learning. **Module (b)** focuses on spatiotemporal representation, where submodule **(b-1)** designs multi-scale time series imaging to represent both short-term variations and long-term recurring patterns, and submodule **(b-2)** constructs multi-granularity spatial graphs to adaptively aggregate nodes while preserving spatial topology. **Module (c)** suggests a Mamba-based enhancement mechanism to fuse temporal and spatial information through cross-domain interaction, refining the learned representations. Finally, **Module (d)** integrates the enhanced features with generates accurate traffic flow predictions.

3.1 Multi-Scale Temporal Representation Module

To effectively capture temporal dependencies across different resolutions, this paper design a multi-scale temporal representation module, which consists of three key components: image construction, temporal attribute extraction, and raw sequence preservation. The temporal features obtained through these three approaches jointly capture both high-level abstractions and fine-grained sequential details for prediction.

3.1.1 Multi-scale time series imaging. To better capture periodic fluctuations and multi-scale temporal dependencies, traffic flow sequences are transformed into two-dimensional heatmaps, where local regions encode short-term variations and global structures reflect long-term trends. The proposed model leverages computer vision techniques to exploit their strength in learning hierarchical and spatially organized representations for more effective temporal feature extraction. The model consists of two main components: scale expansion and image transformation. The former adjusts the time window size P to generate multi-scale sequences, while the latter converts them into structured image representations.

Scale expansion. The time series is expanded across multiple scales by adjusting the time window size P . Specifically, for each time step t , the sequence $(X_t^n)_{Scale_S}$ at the node n and the scale S is constructed by aggregating consecutive observations within non-overlapping windows of length.

$$(X_{t-S \times P+1}^n, \dots, X_t^n)_{Scale_1} \rightarrow \left[(X_{t-S \times P+1}^n + \dots + X_{t-S \times P+S}^n), \dots, (X_{t-S+1}^n + \dots + X_t^n) \right]_{Scale_S}. \quad (2)$$

Furthermore, Eq. (3) defines the computation of each element at the scale S , where S successive values from scale $S-1$ are summed, thereby maintaining the same temporal span while capturing longer-term dynamics.

$$(X_{t-S \times P+1}^n)_{Scale_S} = \left[(X_{t-S \times P+1}^n + \dots + X_{t-S \times P+S}^n), \dots, (X_{t-S+1}^n + \dots + X_t^n) \right]_{Scale_S}. \quad (3)$$

In the process of scale expansion, the number of time steps in the series is directly determined by the scale S . For scale 1, the time series length is P , consisting of P complete data points without missing values. For scales $S > 1$, the time window increases, which reduces the number of data points at each scale to $\frac{P}{S}$, resulting in missing values. To ensure consistent data length across scales and to capture features at different scales, the observation period is extended to compensate for these missing values. Specifically, for scales $S > 1$, a longer time period of $S \times T$ is selected to observe flow variations. The missing entries in the extended period are replaced with zeros, ensuring that each scale's time series retains a consistent length of P .

Image transformation. In the process of time series to image conversion, each time series of a node within a sample window is transformed into a single multi-scale heatmap. Specifically, rows in the heatmap correspond to different temporal scales S , and columns represent time steps. Within each scale, the normalized flow values

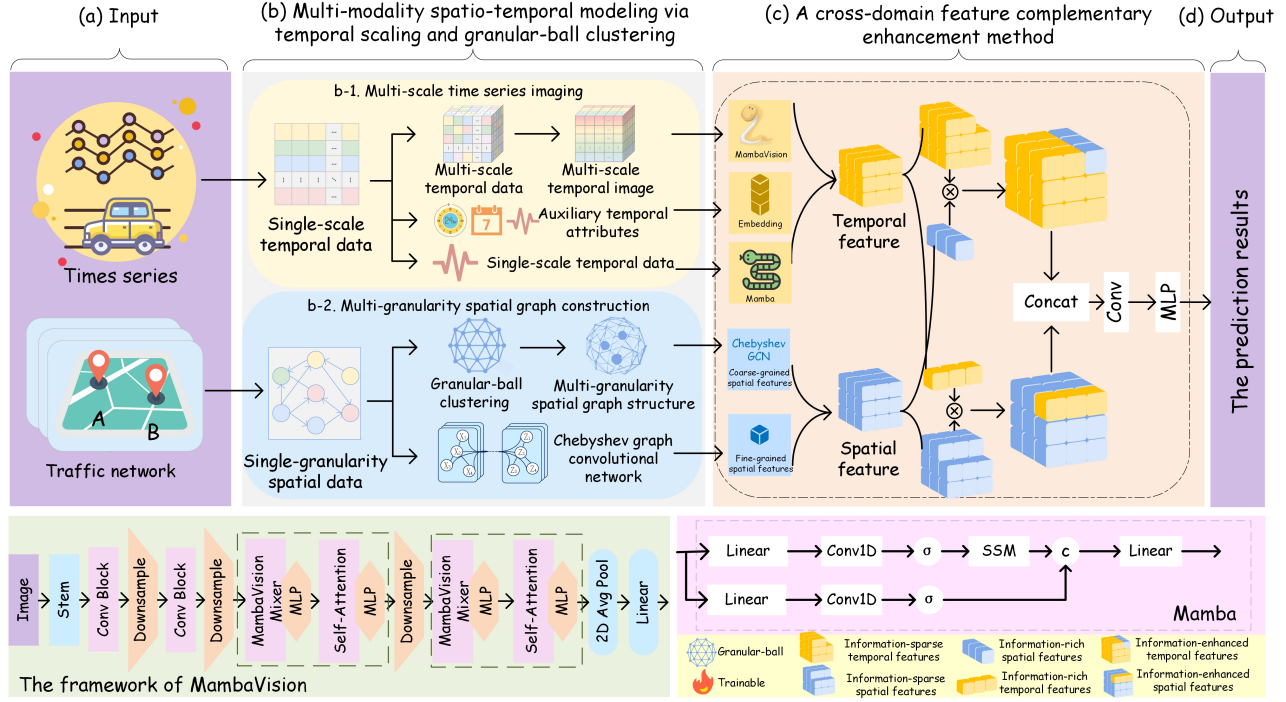


Figure 2: MIGC-CMamba model architecture. Module (a) inputs the raw traffic data; Module (b) is used for spatiotemporal modeling, mainly includes two submodules: (b-1) multi-scale time series imaging and (b-2) multi-granularity spatial graph construction; Module (c) uses Mamba to extract spatiotemporal features and enhances temporal and spatial information via cross-domain supplementation; finally, Module (d) outputs the traffic flow prediction results.

are mapped to pixel intensities, producing a grayscale representation of temporal dynamics. All scales are uniformly allocated along the vertical axis, and each is compressed to an equal height such that the final image has a fixed resolution of $H \times W$. The normalized flow values are mapped to pixel intensities, producing image representations:

$$I^n = \text{ImageTransform}(X^n), \quad (4)$$

the resulting batch of images has dimensions $\mathbb{R}^{B \times N \times H \times W}$, where B is the batch size and (H, W) is the image dimension.

3.1.2 Temporal attribute extraction. Explicit temporal attributes including the hour-of-day H_t , the day-of-week D_t , and the flow volume F_t are extracted and concatenated to form auxiliary temporal features:

$$Z_{attr} = \text{Concat}(H_t, D_t, F_t). \quad (5)$$

3.1.3 Raw sequence preservation. While imaging and temporal feature extraction enrich traffic representation, valuable fine-grained dynamics in the original time series may be lost during transformation. To preserve fine-grained temporal details, the original traffic sequences are retained:

$$X_{raw} \in \mathbb{R}^{T \times N \times C}. \quad (6)$$

3.2 Multi-Granularity Spatial Graph Construction via Granular-Ball Clustering

To preserve spatial topological features while reducing graph complexity and capturing dependencies at multiple granularities, this paper design a spatial graph construction module consisting of two components: a multi-granularity spatial graph generated by granular-ball clustering and the original spatial graph derived from original adjacency matrix. The former adaptively aggregates nodes into granular-balls based on spatial similarity, modeling spatial dependencies at multiple scales, while the latter preserves the fundamental topology of the traffic network. The spatial features obtained from these two graphs can provide complementary representations.

3.2.1 Multi-granularity spatial graph construction. The multi-granularity spatial graph G_c is constructed using granular-ball clustering, which adaptively aggregates traffic network nodes based on spatial similarity. Given the adjacency matrix $A \in \mathbb{R}^{N \times N}$, a similarity matrix Sim_{ij} is first derived.

$$Sim_{ij} = \exp\left(-\frac{A_{ij}^2}{2\sigma^2}\right). \quad (7)$$

Subsequently, nodes are adaptively clustered based on this matrix. Granular-ball clustering enables flexible control of granularity via adjustable clustering radius, effectively modeling multi-scale spatial dependencies in traffic networks.

The core step in granular-ball clustering is the formation of granular-balls. Initially, a node is randomly selected as the center of the first granular-ball, and then new granular-balls are generated around this center based on the similarity between nodes. By iteratively checking for overlaps and similarities between granular-balls, multiple granular-balls are eventually formed, completing the clustering process. To prevent over-aggregation and ensure that multi-scale spatial features are not overly coarse, an adaptive refinement strategy is incorporated into the clustering procedure. Specifically, when the number of generated clusters is lower than the target, large granular-balls are iteratively split into smaller ones if their internal density and purity still satisfy the quality constraints, i.e., granular-ball clustering therefore aims to balance the coverage Cov and the specificity Sp , ensuring compact clusters with distinct boundaries. The calculation formulas are as follows:

$$Cov = \frac{1}{|C|(|C| - 1)} \sum_{i,j \in C, i \neq j} Sim_{ij}, \quad (8)$$

$$Sp = \min_{i,j \in C, i \neq j} Sim_{ij},$$

where $|C|$ represents the total number of samples contained in the cluster C_k . The clustering produces a mapping matrix M where $M_{i,C} = 1$ if a node i belongs to the cluster C .

3.2.2 Original spatial graph construction. The original spatial graph $G_{fine} = (V, A)$ is preserved to maintain fine-grained topological relationships between individual nodes.

3.3 A Cross-Domain Feature Complementary Enhancement Method

Temporal features capture the evolving patterns of traffic flow over time, while spatial features reflect the dependencies between different regions. However, relying solely on one-dimensional local information cannot comprehensively capture the global complexity of the traffic system. To address this limitation, this paper proposes a cross-domain feature supplementation and enhancement method. This method employs an interaction mechanism to complement and enhance temporal and spatial information, effectively overcoming the limitations of each dimension.

3.3.1 Temporal feature extraction and fusion. This module integrates information from three temporal sources: image features, attributes features, and raw sequences features.

MambaVision-based image feature extraction. The MambaVision architecture is employed to extract features from traffic heatmaps, leveraging the combined strengths of convolutional operations and the Mamba framework.

Firstly, the input heatmap I passes through convolutional blocks to extract local spatial features:

$$X_{conv} = \text{CNNBlocks}(I). \quad (9)$$

Next, the convolutional features are then passed through Mamba blocks to model long-range temporal dependencies using selective SSMs:

$$X_{Mamba} = \text{MambaBlocks}(X_{conv}). \quad (10)$$

Finally, global pooling and linear projection generate the image-based temporal features:

$$T_{img} = \text{Linear}(\text{GlobalAvgPool}(X_{Mamba})). \quad (11)$$

This hybrid architecture effectively captures both local variations and global trends in traffic patterns.

Attribute feature embedding. To enhance temporal context modeling, three types of auxiliary attributes are incorporated in addition to the main traffic sequences: hour-of-day, day-of-week, and flow volume. Hour-of-day and Day-of-week capture the periodicity of traffic patterns and are encoded through separate learnable embedding layers, yielding the continuous latent representations $H^{emb} \in \mathbb{R}^{B \times N \times d_h}$ and $D^{emb} \in \mathbb{R}^{B \times N \times d_d}$. The continuous attribute of flow volume is projected into a lower-dimensional space through a linear transformation, enabling the model to capture fine-grained fluctuations in traffic intensity. The resulting feature representations are concatenated and passed through a fusion layer to produce a unified temporal embedding. The specific formulas are as follows:

$$T_{attr} = \text{Linear}([\text{Embed}(H_t), \text{Embed}(D_t), \text{Linear}(F_t)]). \quad (12)$$

Raw sequence feature extraction. To extract fine-grained temporal features from raw traffic sequences, a Mamba-based temporal extractor is employed.

Firstly, the input sequence is first reshaped and linearly projected into a latent space to ensure dimensional consistency for subsequent modeling.

$$X_{proj} = \text{Linear}(\text{Reshape}(X)), \quad X \in \mathbb{R}^{B \times N \times T}. \quad (13)$$

Next, the projected sequence then undergoes temporal modeling through the Mamba mechanism, which employs selective SSMs to adaptively capture relevant temporal patterns:

$$X_{mamba} = \text{MambaBlock}(X_{proj}). \quad (14)$$

Finally, a linear transformation produces the temporal feature representations for downstream spatiotemporal fusion. The process can be formulated as:

$$T_{seq} = \text{Linear}(X_{mamba}) \in \mathbb{R}^{B \times N \times d}. \quad (15)$$

Temporal feature fusion. The extracted image features T_{img} , auxiliary attributes T_{attr} , and sequence features T_{seq} are linearly projected and summed to obtain a unified temporal representation:

$$T = W_{img}T_{img} + W_{attr}T_{attr} + W_{seq}T_{seq}, \quad (16)$$

where W_{img} , W_{attr} , W_{seq} are learnable matrices. This fused feature captures temporal dynamics from all modalities.

3.3.2 Spatial feature extraction and fusion. This module models spatial dependencies at multiple granularities, combining coarse-to-fine and original graph features to generate a comprehensive spatial representation.

Clustered graph features. Spatial dependencies across different granularities are modeled by applying Chebyshev Graph Convolution Networks (ChebGCNs) on the clustered multi-granularity spatial graph.

Firstly, the clustered adjacency matrix $A_{cluster}$ is constructed by aggregating connections between clusters:

$$A_{cluster} = M^T A M. \quad (17)$$

Next, Chebyshev graph convolution is performed on the clustered graph to capture spatial dependencies:

$$S_{cluster} = \sum_{k=0}^K T_k(\tilde{L}_{cluster}) X_{cluster} \Theta_k, \quad (18)$$

where the $\tilde{L}_{cluster}$ denotes the normalized Laplacian of the clustered graph, the T_k represents the k -th order Chebyshev polynomial, the $X_{cluster} = M^T X$ corresponds to the cluster-level features, and the Θ_k are the learnable parameters associated with each polynomial order.

Finally, the cluster-level features are mapped back to the original node space:

$$S_{coarse} = M S_{cluster}. \quad (19)$$

This coarse-to-fine mapping ensures that each node receives information from its assigned cluster while maintaining the original graph structure.

Original graph features. To preserve fine-grained spatial structures, ChebGCNs are directly applied on the original spatial graph constructed from the adjacency matrix. The formulation is given by:

$$S_{fine} = \sum_{k=0}^K T_k(\tilde{L}) X \Theta_k. \quad (20)$$

Spatial feature fusion. The multi-granularity graph features S_{coarse} and the original graph features S_{fine} are fused to obtain the final spatial representation. A simple linear projection followed by element-wise addition is used:

$$S = W_{coarse} S_{coarse} + W_{fine} S_{fine}, \quad (21)$$

where W_{coarse} and W_{fine} are learnable matrices.

3.3.3 Complementary enhancement for spatiotemporal interaction. In traffic flow prediction, traffic at a given time is influenced by the previous time step, with varying influence over time. Enhancing the interaction between spatial and temporal features helps capture these dependencies. However, spatial and temporal features may contain different amounts of information. Spatial features can be sparse, while temporal features often provide crucial supplementary information. Strengthening their interaction fills these gaps and improves feature representation.

First, the channel-wise scaling factors γ_{spa}^c and γ_{temp}^c are computed for the spatial and temporal branches to quantify the information richness of each feature channel.

$$\begin{aligned} \gamma_{spa}^c &= |BN_{spa} \cdot \text{weight}[S]|, \\ \gamma_{temp}^c &= |BN_{temp} \cdot \text{weight}[T]|. \end{aligned} \quad (22)$$

Second, the thresholds for spatial and temporal feature selection are adaptively determined as follows:

$$\begin{aligned} \sigma_{spa} &= \gamma_{spa}^{\min} + \alpha(\gamma_{spa}^{\max} - \gamma_{spa}^{\min}), \\ \sigma_{temp} &= \gamma_{temp}^{\min} + \alpha(\gamma_{temp}^{\max} - \gamma_{temp}^{\min}), \end{aligned} \quad (23)$$

where $\alpha \in [0, 1]$ is a balance coefficient controlling the threshold level.

Third, a spatial channel is retained if its scaling factor γ_{spa}^c exceeds the threshold σ_{spa} ; otherwise, it is enhanced by element-wise multiplication with the temporal features, and vice versa for temporal channels. This interaction supplements information-poor channels and yields the updated spatial and temporal features. The calculation formulas are:

$$\begin{aligned} S' &= \begin{cases} S, & \text{if } \gamma_{spa}^c > \sigma_{spa} \\ S \odot T, & \text{otherwise} \end{cases}, \\ T' &= \begin{cases} T, & \text{if } \gamma_{temp}^c > \sigma_{temp} \\ T \odot S, & \text{otherwise} \end{cases}. \end{aligned} \quad (24)$$

Fourth, the enhanced spatial and temporal features are concatenated to obtain the fused features.

$$F_{fused} = \text{Concat}(S', T'). \quad (25)$$

Fifth, the final traffic flow predictions are generated through a Multi-Layer Perceptron (MLP) head, which maps the fused spatiotemporal representations to the output space, as formulated below:

$$\hat{Y}_t = W_2 \cdot \sigma(W_1 \cdot F_{fused} + b_1) + b_2. \quad (26)$$

Finally, the loss function for MIGC-CMamba is calculated as follows:

$$\mathcal{L} = \frac{1}{T} \sum_{t=1}^T \|\hat{Y}_t - Y_t\|^2, \quad (27)$$

where $\hat{Y}_t$ represents the predicted traffic flow at time step t , and Y_t denotes the corresponding ground-truth value.

4 Experiments

In this section, this paper begins by presenting the datasets and experimental settings, followed by an evaluation of the proposed MIGC-CMamba model for spatiotemporal traffic prediction, aiming to answer the following research questions:

RQ1: Does MIGC-CMamba outperform state-of-the-art baselines across different traffic datasets?

RQ2: How do different components of MIGC-CMamba contribute to its performance?

RQ3: How does the model's performance vary with key hyperparameters, such as image resolution in multi-scale sequence imaging?

RQ4: Can the effectiveness of the proposed framework be demonstrated through visualization results that reveal meaningful spatial partitions and coherent temporal patterns?

4.1 Datasets

Experiments for evaluating the MIGC-CMamba model are performed on four real-world traffic datasets: PeMS03, PeMS-BAY, METR-LA, and JiNan, as detailed in Appendix B.

4.2 Experimental Settings and Evaluation Metrics

The datasets are split into training, validation, and test sets with a ratio of 6:2:2. Single-step traffic flow prediction is performed, with the number of historical time steps P set to 12 and the number

of future time steps Q set to 1. In addition, multi-scale sequence imaging is employed to generate 64×64 temporal representations, and granular-ball clustering is used to construct multi-granularity spatial graphs.

Experiments are conducted on a high-performance platform with an NVIDIA H100 PCIe GPU with 80 GB of memory using CUDA 12.8 and cuDNN, enabling efficient training of large spatiotemporal models like MIGC-CMamba. Three evaluation metrics are used, with definitions provided in Appendix C, and all datasets and code are publicly available at <https://github.com/changwenx/MIGC-CMamba> to ensure reproducibility.

4.3 Baselines (RQ1)

To provide a comprehensive evaluation, this paper selects representative baseline models spanning diverse methodological paradigms in traffic flow prediction, as detailed in Appendix D.

(1) **Statistical methods**: Classical and interpretable baselines that capture linear temporal dependencies.

(2) **RNN-based architectures**: Sequence learning models widely adopted for traffic prediction tasks.

(3) **Graph neural network (GNN)-based architectures**: Models that explicitly incorporate spatial topology, reflecting the mainstream of spatiotemporal modeling.

(4) **Transformer and attention-based models**: Models using attention to capture long-range spatiotemporal dependencies, reflecting recent research trends.

(5) **Mamba-based models**: Emerging approaches designed for efficient modeling of long sequences with improved scalability.

As shown in Table 1, our model achieves the best performance on most datasets and metrics, while remaining competitive elsewhere.

4.4 Ablation Study (RQ2)

To further verify the effectiveness of each component in our model, this paper conduct ablation experiments by progressively removing the proposed modules:

(1) **w/o Multi-Scale Imaging**: removing the multi-scale sequence imaging, where the original traffic time series is directly fed into the model without visual feature extraction.

(2) **w/o Granular-Ball Clustering**: replacing granular-ball clustering with fixed spatial graphs, which ignores the adaptive cognitive advantages of multi-granularity structures.

(3) **w/o Complementary Enhancement**: discarding the cross-domain complementary enhancement module and relying solely on single-domain representations for spatiotemporal modeling.

Table 2: The ablation study on PeMS03. The best performance is highlighted in bold and the second-best is underlined.

Variants	MAE	RMSE	MAPE (%)	LOSS
w/o Multi-Scale Imaging	28.19	38.68	23.41	0.20
w/o Granular-Ball Clustering	<u>28.04</u>	<u>38.59</u>	<u>23.05</u>	0.20
w/o Complementary Enhancement	28.22	38.71	23.41	<u>0.19</u>
Full model (Ours)	13.35	19.98	14.29	0.06

As shown in Table 2, removing any of the proposed modules leads to a clear performance drop, demonstrating that each part plays an indispensable role. Among them, multi-scale imaging has the most significant impact on capturing trend and periodic patterns, granular-ball clustering improves spatial adaptability, and complementary enhancement further strengthens cross-domain spatiotemporal interactions.

4.5 Parameter Sensitivity Study (RQ3)

To assess the effect of image resolution in the multi-scale sequence imaging module, this paper performs a parameter sensitivity analysis by testing image sizes of 4×4 , 8×8 , 16×16 , 32×32 , 64×64 , 128×128 .

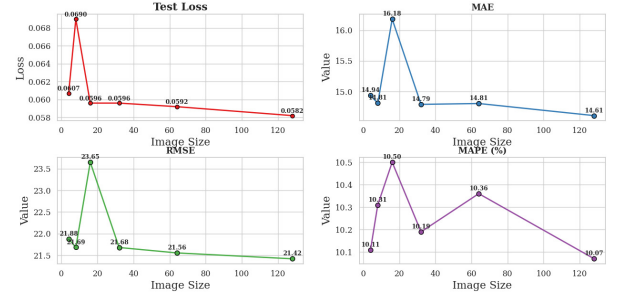


Figure 3: The impact of image resolution on test loss and prediction performance.

As shown in Figure 3, overly low resolutions (e.g., 4×4) lose fine-grained spatial-temporal details and degrade accuracy, while very high resolutions (e.g., 128×128) offer marginal gains at substantial computational cost. Overall, 64×64 provides the best balance between performance and efficiency.

4.6 Visualization (RQ4)

Visualization helps interpret the model by revealing the patterns it captures. This paper evaluates the spatial feature extraction strategy by examining whether clustering produces meaningful partitions and groups nodes with similar traffic dynamics. The analysis examines two aspects: network topology and temporal consistency within clusters.

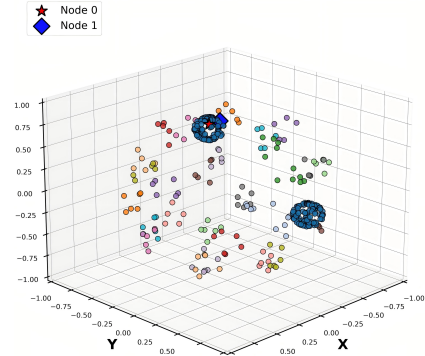


Figure 4: The clustering results on the traffic network.

Table 1: The performance comparison on PeMS03, PeMS-Bay, METR-LA, JiNan. The best performance is highlighted in bold and the second-best is underlined.

Models	PeMS03			PeMS-Bay			METR-LA			JiNan		
	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)
HA	31.59	52.40	33.79	4.51	7.83	15.62	4.68	8.24	17.41	13.25	22.86	46.84
ARIMA	35.32	47.51	32.67	4.01	7.21	13.11	4.16	7.96	14.81	14.90	26.23	48.54
VAR	23.64	38.24	24.53	1.76	3.15	3.61	4.43	7.82	13.01	12.57	20.61	40.55
LSTM	20.97	47.60	20.77	3.03	5.74	9.82	3.26	6.57	11.83	11.31	19.41	38.01
GRU	20.20	35.09	20.89	3.02	5.74	9.72	3.26	6.54	11.72	12.11	19.40	36.95
STGCN	17.54	30.41	17.33	2.89	5.88	5.16	3.23	7.54	9.83	9.42	16.10	36.41
ASTGCN	17.24	29.51	17.23	2.72	5.49	4.83	3.36	7.06	9.34	10.08	17.21	38.92
DCRNN	18.01	30.32	18.32	1.40	2.96	3.21	2.78	5.39	7.31	11.41	20.55	41.75
HSTGNN	14.55	<u>24.18</u>	13.76	1.32	2.78	<u>2.70</u>	2.68	5.36	<u>6.51</u>	9.65	15.65	38.62
STDN	17.06	28.54	16.54	2.61	2.87	3.65	2.65	5.24	6.56	<u>8.39</u>	14.58	32.29
PDFormer	14.92	25.36	15.85	1.33	2.84	2.78	2.79	5.46	7.79	8.52	<u>13.56</u>	<u>32.26</u>
DTRformer	14.51	25.46	<u>14.95</u>	1.32	2.80	2.76	2.67	5.12	7.05	8.61	14.24	34.56
MDPM	<u>14.33</u>	24.83	15.55	1.35	<u>2.74</u>	2.83	2.71	<u>5.11</u>	7.02	9.64	13.65	33.42
ST-Mamba	15.29	27.31	15.15	1.31	2.88	2.93	2.67	5.18	6.62	9.12	13.58	33.62
ST-MambaSync	15.32	27.48	15.21	<u>1.31</u>	2.76	2.71	<u>2.64</u>	5.05	6.80	9.21	13.61	33.59
Ours	13.67	20.21	15.20	1.20	1.81	2.30	2.40	4.15	5.87	8.00	12.98	32.19
Improvement (%)	4.60	16.41	-	8.40	33.94	14.81	9.09	18.79	9.83	4.65	4.28	0.21

As shown in Fig. 4, the granular-ball clustering generates compact and spatially coherent regions while effectively preserving the overall topology, thereby reducing graph complexity. To further illustrate its spatial coherence, two nodes within the same cluster are selected for analyzing their traffic flow patterns.

**Figure 5: The traffic flow trajectories of two nodes within the same cluster.**

Fig. 5 illustrates that nodes within the same cluster exhibit consistent temporal patterns, including synchronized peak hours, fluctuation amplitudes, and overall trends, confirming that the clustering strategy effectively groups nodes with similar traffic dynamics and underlying functional correlations within the network.

5 Conclusion and Future Work

This paper presents a novel MIGC-CMamba framework that unifies multi-scale sequence imaging, granular-ball-based multi-granularity spatial graph construction, and cross-domain complementary enhancement for traffic flow prediction. The framework encodes temporal sequences as images to capture both local and global temporal trends, employs granular-ball clustering to enable adaptive multi-granularity spatial modeling, and further enhances spatial-temporal interactions to overcome the inherent limitations of static graph structures and single-scale modeling methods. The detailed algorithm of MIGC-CMamba is provided in Appendix E. Experiments on real-world datasets demonstrate its consistent superiority over strong baselines, while future work will focus on improving efficiency, scalability, and sensitivity analysis.

Acknowledgments

The paper was supported by grants from the National Natural Science Foundation of China (62272284, 62473241, 12201518), the Central Government Guides Local Science and Technology Innovation (YDZJSX2024D015), the Science and Technology Cooperation and Exchange Special Projects of Shanxi (Research on Privacy Enhancement Technology for Trusted Data Space Based on Quantum Multi-Granularity Group Consensus), the Top Young Talents of Shanxi “Three Jin” Talents Program, the Sichuan Science and Technology Program (2025ZNSFSC0806), the Wenying Young Scholars of Shanxi University, and the Project of Science and Technology Research Program of Chongqing Education Commission (KJZD-K20250020).

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A Related Work

This section reviews related studies on traffic flow prediction from three perspectives: traditional traffic flow prediction methods, multi-scale time series imaging, and multi-granularity spatial graph construction based on granular-balls. These three research directions collectively reflect the evolution of traffic prediction from statistical modeling to deep learning approaches capable of capturing complex temporal and spatial dependencies.

A.1 Traditional Traffic Flow Prediction Methods

Research on traffic flow prediction has transitioned from traditional statistical and machine learning approaches to modern deep learning and structured sequence models, increasing the capacity to represent complex temporal dynamics and their interactions across the spatial network. Early research on time series prediction rely on statistical models such as Autoregressive Integrate Moving Average (ARIMA) and Vector Autoregression (VAR) [19], which are interpretable and parameter-efficient but limited in capturing nonlinear patterns and long-term dependencies. To address these limitations, machine learning methods including support vector regression and ensemble learning are explored [1], though they depend on handcrafted features and cannot automatically model temporal dynamics. With the rise of deep learning, RNNs and their variants

such as LSTM and Gated Recurrent Unit (GRU) [7] enable non-linear sequence modeling but suffer from sequential computation and vanishing gradients. Convolutional architectures such as Convolutional Neural Networks (CNNs) and Temporal Convolutional Networks (TCNs) [16] improve parallelism and receptive fields, yet remain insufficient for capturing global dependencies. Transformer-based models [15] further address long-range dependencies via self-attention, but their quadratic complexity limits scalability to ultra-long sequences. Recently, structured SSMs emerge as efficient alternatives for long-term dependency modeling [22], and Mamba and its variants [5] demonstrate strong scalability and effectiveness on ultra-long sequences. Compared with traditional approaches, incorporating multi-scale temporal modeling and multi-granularity spatial representation offers a more comprehensive understanding of traffic dynamics by jointly capturing both fine-grained variations and global structural patterns. Despite these advances, effectively modeling multi-scale temporal dependencies and multi-granularity spatial correlations remains a significant challenge.

A.2 Multi-Scale Time Series Imaging

Traffic flow exhibits complex temporal dynamics, including short-term fluctuations and long-term periodicity, which are difficult to capture using single-scale sequence modeling. Multi-scale modeling has proven effective in capturing temporal dependencies at different resolutions, allowing both fine- and coarse-grained patterns to be represented [24, 26]. Furthermore, transforming one-dimensional traffic sequences into two-dimensional image-like representations enables the exploitation of spatially structured feature extraction and the application of powerful neural network operators, such as CNNs, to capture richer temporal correlations [17]. Despite these advantages, existing imaging-based methods often suffer from high computational costs and struggle to preserve fine-grained temporal information, limiting their ability to efficiently capture multi-scale temporal dependencies.

A.3 Multi-Granularity Spatial Graph Construction Based on Granular-balls

Spatial dependencies in traffic networks exhibit heterogeneity across multiple granularities, from individual nodes and road segments to broader regional clusters, making effective multi-granularity modeling essential for accurate prediction [34]. Existing methods often rely on manually designed partitions or fixed rules, which are inflexible and lack a unified theoretical foundation. Granular computing (GC) [32] provides a natural paradigm for organizing information into hierarchical granules, enabling efficient and scalable multi-granularity representation. Early GC-based methods treated granules mainly as auxiliary feature-processing tools without fundamentally modifying underlying classifiers. To improve integration, Xia et al. [27] introduced granular-ball, leveraging the symmetry and simplicity of spheres to better incorporate granules into classifiers. Subsequent studies refined granular-ball generation by considering factors such as radius, sample size [33], and distribution-based methods for unlabeled data [28], enhancing robustness and

adaptive multi-granularity representation. Granular-ball clustering offers an adaptive and scalable approach to multi-granularity modeling, capable of automatically determining cluster boundaries based on data distribution rather than fixed rules. It achieves a balance between global structural integrity and local detail preservation, enhancing robustness against noise and uneven data density. Despite these advances, applying granular-ball-based modeling to traffic flow prediction remains underexplored, and effectively capturing multi-granularity spatial dependencies in large-scale traffic networks continues to pose a significant challenge.

B Dataset

This paper evaluates the MIGC-CMamba model on four real-world traffic datasets: PeMS03, PeMS-BAY, METR-LA, and JiNan. PeMS03 and PeMS-BAY reflect traffic flow on different highway networks in California, METR-LA represents complex urban traffic conditions in Los Angeles, and JiNan captures inner-city traffic dynamics in China. Together, these datasets provide representative benchmarks across diverse traffic scenarios, as summarized in Table 3.

Table 3: The summary of traffic datasets

Datasets	Nodes	Time steps	Time ranges	Days
PeMS03	358	26,208	09/2018–11/2018	91
PeMS-BAY	325	52,116	01/2017–05/2017	151
METR-LA	207	34,272	03/2012–06/2012	122
JiNan	406	8,928	08/2016–09/2016	31

C Evaluation Metrics

This paper adopt three evaluation metrics: mean absolute error (MAE), root mean squared error (RMSE), and mean absolute percentage error (MAPE). Lower values indicate superior predictive performance. The definitions of the three metrics are given as follows.

$$\begin{aligned}
 MAE &= \frac{1}{m} \sum_{i=1}^m |\hat{Y}_i - Y_i|, \\
 RMSE &= \sqrt{\frac{1}{m} \sum_{i=1}^m (\hat{Y}_i - Y_i)^2}, \\
 MAPE &= \frac{100}{m} \sum_{i=1}^m \left| \frac{\hat{Y}_i - Y_i}{Y_i} \right|,
 \end{aligned} \tag{28}$$

where m denotes the number of predicted values, $\hat{Y}_i$ is the predicted traffic flow, and Y_i is the ground truth.

D Baselines

Representative baseline models covering various methodological paradigms and recent advances in traffic flow prediction are selected for evaluation, ensuring a comprehensive comparison across statistical, machine learning, and deep learning approaches from recent years.

Statistical methods.

HA [6] predicts future traffic flow using historical averages from corresponding time slots, offering a simple yet effective baseline.

ARIMA [13] captures linear temporal dependencies through autoregressive and moving average components.

VAR [20] extends univariate autoregressive models to multivariate settings, capturing linear interdependencies among multiple time series.

RNN-based architectures.

LSTM [31] uses gated memory units to capture the long-term temporal dependencies, effectively modeling sequential traffic flow data while alleviating the vanishing gradient problem.

GRU [4] is a streamlined LSTM variant that achieves similar predictive performance with fewer parameters and lower computational cost.

GNN-based architectures.

STGCN [30] is a fully convolutional deep learning framework based on graph convolutions that efficiently captures spatiotemporal dependencies for accurate traffic flow prediction.

ASTGCN [8] is an attention-based spatiotemporal graph convolutional network that models recent, daily, and weekly traffic dependencies, using spatial-temporal attention and convolutions to capture dynamic spatiotemporal correlations for accurate traffic flow prediction.

DCRNN [35] employs RNN-based cross networks for enhanced feature interaction, adopts a novel partial parameter sharing architecture, and effectively captures task correlations to improve multi-task learning efficiency.

STDN [2] employs a dynamic graph with spatiotemporal embeddings, trend-seasonality decomposition, and an encoder-decoder network for accurate and efficient traffic prediction.

Transformer and attention-based models.

HSTGNN [25] is a hybrid spatiotemporal graph neural network that captures heterogeneous temporal patterns via frequency-based decoupling and dual-channel learning, while modeling dynamic spatial correlations through hybrid graph learning, enabling accurate long-term traffic prediction.

PDFormer [11] leverages spatiotemporal attention and progressive feature decomposition to efficiently capture long-range dependencies, achieving accurate traffic flow prediction.

DTRformer [3] is a transformer-based framework that captures spatiotemporal correlations in traffic data by combining dynamic trend representations and static graph information through cross spatial-temporal attention for accurate traffic flow prediction.

MDPM [37] uses a domain-shared encoder to capture universal spatiotemporal patterns, a gated fusion mechanism for cross-domain alignment, and domain-specific prompt vectors for dataset-specific traffic features.

Mamba-based models.

ST-Mamba [22] captures long-range spatiotemporal dependencies in traffic flow without graph modeling by integrating spatial and temporal data via a selective state space framework, improving both prediction accuracy and computational efficiency.

ST-MambaSync [22] is a transformer-based traffic prediction model that integrates the ST-Mamba block, combining attention with ResNet to handle long sequences efficiently.

E Algorithm

The algorithm proposed in this paper is as follows.

Algorithm 1 The algorithm of MIGC-CMamba.

Require: The historical traffic flow data $X \in \mathbb{R}^{T \times N \times C}$

The adjacency matrix $A \in \mathbb{R}^{N \times N}$

The number of clusters K , Balance coefficient α

The prediction horizon T'

Ensure: The predicted traffic flow $\hat{Y} \in \mathbb{R}^{T' \times N}$

```

1: // Multi-Scale Temporal Representation
2: for each node  $n = 1$  to  $N$  do
3:   Apply scales expansion using the Eq. (2) and Eq. (3)
4:   Transform time series to image:  $I^n = \text{ImageTransform}(X^n)$ 
5: end for
6: Extract temporal attributes:  $Z_{attr} = \text{Concat}(H_t, D_t, F_t)$ 
7: Preserve the raw sequence:  $X_{raw} = X$ 
8: // Multi-Granularity Spatial Graph Construction
9: Compute the similarity matrix:  $Sim_{ij} = \exp\left(-\frac{A_{ij}^2}{2\sigma^2}\right)$ 
10: Apply the granular-ball clustering to generate the mapping matrix  $M$ 
11: Construct the clustered graph:  $A_{cluster} = M^T A M$ 
12: Preserve the original graph:  $\mathcal{G}_{fine} = (V, A)$ 
13: // Temporal Feature Extraction
14: Extract image features:  $T_{img} = \text{MambaVision}(I)$ 
15: Extract attribute features:  $T_{attr} = \text{Embedding}(Z_{attr})$ 
16: Extract sequence features:  $T_{seq} = \text{MambaBlock}(X_{raw})$ 
17: Fuse temporal features:  $T = W_{img}T_{img} + W_{attr}T_{attr} + W_{seq}T_{seq}$ 
18: // Spatial Feature Extraction
19: Extract clustered graph features:  $S_{cluster} = \text{ChebGCN}(A_{cluster}, X)$ 
20: Map to the node space:  $S_{coarse} = M S_{cluster}$ 
21: Extract original graph features:  $S_{fine} = \text{ChebGCN}(A, X)$ 
22: Fuse spatial features:  $S = W_{coarse}S_{coarse} + W_{fine}S_{fine}$ 
23: // Cross-domain Complementary Enhancement
24: Compute scaling factors  $\gamma_{spa}$  and  $\gamma_{temp}$  using Eq. (15)
25: Compute thresholds  $\sigma_{spa}$  and  $\sigma_{temp}$  using Eq. (16)
26: for each channel  $c = 1$  to  $d$  do
27:   if  $\gamma_{spa}^c > \sigma_{spa}$  then
28:      $S' = S$ 
29:   else
30:      $S' = S \odot T$ 
31:   end if
32:   if  $\gamma_{temp}^c > \sigma_{temp}$  then
33:      $T' = T$ 
34:   else
35:      $T' = T \odot S$ 
36:   end if
37: end for
38: Fuse features:  $F_{fused} = \text{Concat}(S', T')$ 
39: // Prediction and Optimization
40: Generate predictions:  $\hat{Y} = \text{MLP}(F_{fused})$ 
41: Compute the loss:  $\mathcal{L} = \frac{1}{T} \sum_{t=1}^T \|\hat{Y}_t - Y_t\|^2$ 
42: Update model parameters via backpropagation
43: return  $\hat{Y}$ 

```
